

RITA AI / AnythingLLM Change Report

Prepared for RITA rebrand and chart/report generation improvements. This document summarizes the main implementation changes made to the AnythingLLM codebase, including chart generation, matplotlib support, PDF chart reports, RITA capability toggles, admin settings, file download improvements, Docker changes, and new dependencies.

Generated: 5 June 2026

Executive Summary

- Improved chart generation by adding ECharts support to the existing chart renderer and backend chart plugin.
- Added a new matplotlib agent skill that can run Python matplotlib scripts and return PNG/JPG/SVG/PDF chart files.
- Added a new combined create-chart-pdf-report tool so RITA can generate chart images and embed them directly into a PDF report.
- Added a Report Builder option in the chat tools menu for guided PDF/chart report requests.
- Added a RITA capability registry in system_settings so features can be enabled or disabled from an admin settings panel.
- Improved generated file download/preview behavior for chart and report outputs.
- Added Docker dependencies for Python/matplotlib and env settings for auto-approved agent skills.
- Fixed model/tool-call parsing issues caused by smart quotes in tool JSON.

New Installations / Dependencies

Frontend

- Added echarts ^5.5.1 to render richer and more flexible charts in the chat UI.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/package.json

Added the echarts dependency.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/package-lock.json

Updated lockfile for echarts installation.

Docker Runtime

- Added python3 and python3-matplotlib in Docker so agent tools can generate matplotlib charts inside the container.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/docker/Dockerfile

Installs python3 and python3-matplotlib in the Docker image.

Server Dependency Fix

- Updated server package metadata/lock files to fix the zod/v3 package export error encountered during Docker startup.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/package.json

Updated server dependency metadata.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/package-lock.json

Updated server lockfile after dependency change.

Chart Generation Improvements

ECharts Renderer

- The chart UI can now render ECharts option JSON while keeping support for older chart payloads.
- This makes generated charts more flexible than the previous simpler chart rendering path.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/components/WorkspaceChat/ChatContainer/ChatHistory/Chartable/index.jsx

Added ECharts import/rendering and support for chart option payloads such as option, echartsOption, or chartSpec.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibitat/plugins/rechart.js

Updated backend chart plugin to accept optional ECharts option JSON and send it to the frontend chart renderer.

Matplotlib Chart Agent Skill

- Added a create-matplotlib-chart tool for high quality Python/matplotlib charts.
- The tool accepts filename, format, and Python code, then saves the generated chart as a downloadable file.
- Supported formats include PNG, JPG/JPEG, SVG, and PDF.
- The tool sanitizes smart quotes/code fences and uses a non-interactive matplotlib backend.
- Python command detection now skips the Windows Microsoft Store python alias error and tries other candidates.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibitat/plugins/create-files/python/create-matplotlib-chart.js

New matplotlib chart generation skill. Runs Python safely in a temporary directory and saves generated chart files.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibitat/plugins/create-files/index.js

Registered create-matplotlib-chart in the create-files agent.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/pages/Admin/Agents/CreateFileSkillPanel/index.jsx

Added Create File skill toggle entry for Matplotlib Charts.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/locales/en/common.js

Added UI title/description strings for Matplotlib Charts.

PDF Chart Report Tool

- Added create-chart-pdf-report as a combined chart and PDF tool.
- This solves the previous problem where create-pdf-file could only make markdown/text PDF reports and the agent wrote a fallback note saying chart images could not be embedded.
- The new tool receives structured chart specs, generates chart PNGs with matplotlib, embeds the PNGs into a PDF using pdf-lib, and returns one downloadable PDF.
- Supported chart types: bar, horizontal_bar, line, pie, donut.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibitat/plugins/create-files/pdf/create-chart-pdf-report.js

New combined tool that generates chart images and embeds them into a PDF report.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibitat/plugins/create-files/index.js

Registered create-chart-pdf-report in the create-files agent.

C:/Users/mohdb/Desktop/Kerja

Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/pages/Admin/Agents/CreateFileSkillPanel/index.jsx

Added Create File skill toggle entry for Chart PDF Reports.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/locales/en/common.js

Added UI title/description strings for Chart PDF Reports.

Report Builder in Chat Tools Menu

- Added a guided Report Builder tab inside the chat plus/tools menu.
- Users can choose report focus, chart types, output format, and extra instructions.
- The generated prompt now tells the agent to use create-chart-pdf-report for chart PDFs instead of the old create-pdf-file fallback.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/components/WorkspaceChat/ChatContainer/PromptInput/ToolsMenu/Tabs/ReportBuilder.jsx

New Report Builder UI for guided PDF/chart report prompts.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/components/WorkspaceChat/ChatContainer/PromptInput/ToolsMenu/index.jsx

Registered the Report Builder tab and connected it to RITA capability settings. Also allows inputs in the tools menu to keep focus.

RITA Capability Registry / Admin Feature Toggles

- Added rita_capabilities as a JSON setting in system_settings.
- Default capabilities are report_builder, chart_generator, and pdf_report_generator.
- Admin Agents page now has a RITA Capabilities panel to enable/disable these features.
- The chat tools menu reads RitaCapabilities from system settings and hides Report Builder when disabled.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/models/systemSettings.js

Added defaultRitaCapabilities, validation, public/supported field registration, currentSettings exposure, and ritaCapabilities helper.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/endpoints/admin.js

Exposes rita_capabilities through /admin/system-preferences-for.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/pages/Admin/Agents/index.jsx

Added RITA Capabilities admin panel and saves rita_capabilities through system preferences.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/components/WorkspaceChat/ChatContainer/PromptInput/ToolsMenu/index.jsx

Reads RitaCapabilities and controls whether Report Builder appears.

Agent Tool Calling and Download Fixes

- Improved parsing of tool JSON when models output smart quotes or imperfect tool-call syntax.
- Improved generated file card preview/download behavior so generated chart/report files show and download more reliably.

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Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/utils/agents/aibatat/providers/helpers/untooled.js

Improved parser robustness for malformed/smart-quoted tool call JSON.

C:/Users/mohdb/Desktop/Kerja

Hypepresso/PKNS-RITA/RITA/rita-hypepresso/frontend/src/components/WorkspaceChat/ChatContainer/ChatHistory/FileDownloadCard/index.jsx

Improved generated file card display/download handling for chart/report files.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/endpoints/agentFileServer.js

Improved generated file serving/download timing and access handling.

Environment / Docker Changes

- Added PYTHON_BIN hint to env examples for selecting the Python executable used by matplotlib tools.
- Added create-chart-pdf-report to auto-approved skills to reduce repeated approval prompts.
- Docker Compose was adjusted during local testing so the app can run without conflicting with an existing service on port 3001.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/docker/.env.example

Documents PYTHON_BIN and AGENT_AUTO_APPROVED_SKILLS including create-chart-pdf-report.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/server/.env.example

Documents PYTHON_BIN and AGENT_AUTO_APPROVED_SKILLS including create-chart-pdf-report.

C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/docker/docker-compose.yml

Local compose port mapping changed during setup/testing.

Note: docker/.env was also updated locally to auto-approve create-matplotlib-chart, create-pdf-file, and create-chart-pdf-report. Secrets are intentionally not shown in this report.

How to Test

Rebuild and Start Docker

- cd "C:/Users/mohdb/Desktop/Kerja Hypepresso/PKNS-RITA/RITA/rita-hypepresso/docker"
- docker compose down
- docker compose build anything-llm
- docker compose up

Test Matplotlib Chart

@agent create a matplotlib PNG bar chart for Jan 10, Feb 20, Mar 15.

Test PDF Chart Report

@agent create a PDF chart report for Jan 10, Feb 20, Mar 15. Include a bar chart, short executive summary, key finding, and recommendation.

Test Admin Capability Toggle

- Go to Settings -> Agents.
- Use the RITA Capabilities panel to disable Report Builder.
- Open chat plus/tools menu and confirm Report Builder is hidden. Re-enable it and confirm it returns.

Verification Already Performed

- Frontend targeted ESLint passed for modified UI files.
- Backend syntax checks passed for modified/new Node files.
- Frontend production build passed with existing Vite warnings only.
- Full local chart-PDF runtime smoke test could not complete on the Windows host because no usable Python command was exposed in the sandbox. Docker should support it because python3 and python3-matplotlib were added to the Dockerfile.